

Notes: Greenstone, Oyer, and Vissing-Jorgensen (QJE, 2006)
Mandated Disclosure, Stock Returns, and the 1964 Securities Acts
Amendments

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1 Big picture

- **Research question.** Do mandatory disclosure requirements increase shareholder value? More specifically, what happened when the 1964 Securities Acts Amendments extended major disclosure rules from exchange-listed firms to large Over-the-Counter firms?
- **Main idea.** If disclosure reduces insiders' ability to divert resources, trade on private information, avoid scrutiny, or otherwise manage firms for private benefits, then outside shareholders should value the imposition of disclosure rules.
- **Empirical setting.** The paper studies the 1964 Amendments, which applied several disclosure requirements to large OTC firms that had previously faced weaker disclosure obligations than NYSE/AMEX firms.
- **Treatment variation.** The law affected OTC firms differently. Some firms went from zero to four disclosure requirements, some went from two to four, and smaller OTC firms were less likely to be covered. NYSE/AMEX firms were already subject to all four requirements and serve as comparison firms.
- **Core outcome.** The main outcome is abnormal excess stock returns around the debate, passage, and initial enforcement of the law.
- **Main finding.** The OTC firms most affected by the law had positive abnormal excess returns. In the event study around SEC registration, affected OTC firms earned roughly 3.5 percent cumulative abnormal excess returns.
- **Total Period 1 effect.** Between the initial legislative proposal and the effective implementation period, the most affected OTC firms earned abnormal excess returns of about 11.5 to 22.1 percent relative to matched listed firms in the preferred specifications.
- **Validation.** In Period 2, after the key uncertainty about the law and compliance had been resolved, the paper finds no comparable abnormal excess returns. This supports the interpretation that Period 1 returns reflect the disclosure shock.
- **Real outcomes.** The most affected OTC firms also improved operating performance from 1962 to 1966 relative to unaffected listed firms, especially in sales growth and net income growth.
- **Economic takeaway.** Mandatory disclosure can raise outside shareholder value by reducing agency problems and pushing managers toward shareholder-value maximization, although the paper cannot fully separate welfare gains from transfers from insiders to outside shareholders.

2 Incremental contribution of the paper

- **From cross-country correlations to a U.S. quasi-experiment.** Much of the investor-protection literature uses cross-country variation. This paper studies a major U.S. regulatory change with firm-level treatment heterogeneity.
- **A cleaner comparison group.** The 1964 Amendments affected large OTC firms but did not affect NYSE/AMEX firms, because listed firms already complied with the full disclosure regime. This creates a natural untreated benchmark.

- **Hand-collected OTC return data.** OTC firms were not in CRSP for this period, so the authors build a CRSP-like weekly return file for 1,196 OTC firms from Barron’s and other historical sources.
- **Multiple layers of evidence.** The paper does not rely on one test. It combines a firm-level event study, cross-OTC-group comparisons, four-factor matched portfolio returns, a Period 2 placebo-style validation, and operating-performance tests.
- **Treatment intensity.** The 0–4, 2–4, 0–0, and 2–2 group structure turns the law into a test of whether firms more affected by disclosure requirements experienced larger stock-market gains.
- **Market-value interpretation.** The estimated abnormal returns imply sizable shareholder-value gains, around \$3.2 to \$6.2 billion in 2005 dollars for the 0–4 firms in the sample.
- **Governance mechanism.** The results support the view that disclosure rules can reduce insider diversion or increase managerial focus on shareholders. The paper is careful that stock-market gains need not equal social welfare gains.

3 Institutional setting and treatment variation

3.1 Disclosure regime before 1964

Before the 1964 Amendments, U.S. securities regulation created a sharp difference between exchange-listed firms and OTC firms.

- **Listed firms.** NYSE and AMEX firms were subject to four major disclosure requirements: SEC registration, periodic reporting, proxy disclosure, and insider ownership/trading disclosure.
- **Some OTC issuers.** OTC firms that had made sufficiently large public offerings after 1936 were already required to register and file periodic reports, but they were exempt from proxy and insider-disclosure rules.
- **Other OTC issuers.** OTC firms that had not triggered the post-1936 registration regime could be free from all four disclosure requirements, even if they were economically large.
- **Regulatory anomaly.** The pre-1964 system created a “double standard” of investor protection: listed shareholders received much more disclosure than many OTC shareholders.

Interpretation. The paper views this uneven regulatory regime as useful empirical variation. The reform did not simply change disclosure for everyone; it changed disclosure for specific categories of OTC firms.

3.2 The 1964 Securities Acts Amendments

The Amendments extended disclosure requirements to large OTC firms.

- **Coverage rule.** OTC firms with at least \$1 million in assets and enough shareholders were required to register with the SEC and comply with the broader disclosure regime.

- **Main new requirements.** Covered firms had to register securities, file regular financial reports, provide proxy statements, and disclose insider holdings and trades.
- **Timing.** Period 1 runs from January 1, 1963 to November 15, 1965, covering proposal, passage, and the revelation of which firms would comply. Period 2 runs from November 15, 1965 through the end of 1966, when little new information about the law was expected.
- **Event-study timing.** Because initial filing dates varied by fiscal year end, the paper can also study stock returns around firm-specific announcements that firms had begun complying.

Interpretation. The law generates both cross-sectional variation in treatment intensity and time-series variation in when information about the law and compliance became public.

3.3 OTC treatment groups

The authors classify OTC firms using information available at the start of 1963, before serious legislative action.

- **0–4 firms.** Firms above the size thresholds that were not filing with the SEC before the Amendments. These firms were expected to move from zero to four disclosure requirements.
- **2–4 firms.** Firms above the size thresholds that were already filing periodic reports. These firms were expected to move from two to four disclosure requirements.
- **0–0 firms.** Smaller nonfiling OTC firms. These firms had low expected compliance probability but potentially large disclosure changes if they grew into coverage.
- **2–2 firms.** Smaller OTC firms already filing periodic reports. These firms had low expected compliance probability and relatively small additional disclosure exposure.
- **4–4 firms.** NYSE/AMEX listed firms. These firms were already subject to all four requirements and are unaffected comparison firms.

Treatment ranking. If investors valued disclosure, the 0–4 firms should experience the largest positive treatment effect, the 2–2 firms the smallest, and the 2–4 and 0–0 firms intermediate effects.

4 Data and measurement

4.1 Hand-built OTC stock-return data

A central data contribution is the construction of a weekly OTC return panel.

- **Starting sample.** The OTC sample begins with firms listed in Barron’s OTC tables in the first week of 1963.
- **Final sample.** After sample restrictions, the main OTC return file contains 1,196 OTC firms.
- **Return period.** Weekly returns are constructed from 1963 through 1966.
- **Price source.** OTC bid and ask prices are collected from weekly Barron’s issues. Returns use the midpoint of bid and ask prices.

- **Other sources.** Dividends, splits, mergers, name changes, liquidations, SEC filing status, and accounting variables come from sources such as S&P Annual Dividend Record, Moody’s manuals, SEC directories, and SEC News Digest.
- **Why hand collection matters.** OTC firms were not available in CRSP during this period, so the paper creates the historical return data needed for the test.

Interpretation. Without the hand-built OTC data, the 1964 Amendments could not be evaluated with the same kind of return-based empirical design used for listed firms.

4.2 NYSE/AMEX comparison groups

The authors use CRSP and COMPUSTAT to construct comparison portfolios of unaffected listed firms.

- **4–4 status.** NYSE/AMEX firms already complied with all four disclosure requirements before the 1964 Amendments.
- **Matching variables.** The comparison groups are matched to OTC firms on market capitalization and book-to-market equity.
- **Reason for matching.** Average returns differ systematically by size and book-to-market, so the comparison groups should resemble the OTC portfolios on those dimensions.
- **Separate comparison groups.** The authors construct separate 4–4 comparison portfolios for the OTC groups.

Interpretation. The 4–4 groups are not treated by the law, so abnormal returns of affected OTC groups relative to matched 4–4 firms are interpreted as the market’s valuation of the new disclosure requirements.

4.3 Return periods and compliance events

The paper uses three key return windows.

- **Period 1.** January 1, 1963 to November 15, 1965. This is the main treatment window, covering legislative proposal, passage, and initial compliance information.
- **Period 1a.** January 1, 1963 to August 24, 1964. This covers the debate and passage of the law.
- **Period 1b.** August 24, 1964 to November 15, 1965. This covers initial compliance filings and the revelation of which firms registered.
- **Period 2.** November 15, 1965 to December 31, 1966. This is the validation window; the main new information about the law and compliance should already have been revealed.
- **Event window.** The firm-level event window runs from eight weeks before to one week after the SEC News Digest announcement of initial compliance.

Interpretation. If the design is valid, effects should be concentrated in Period 1 and around compliance events, not in Period 2.

4.4 Operating-performance measures

The paper also studies whether stock-market gains correspond to real operating changes.

- **Net income growth.** Change in net income from 1962 to 1966, scaled by 1962 market capitalization.
- **Sales growth.** Percentage change in sales from 1962 to 1966.
- **Comparison.** 0–4 and 2–4 firms are compared to 4–4 listed firms, controlling for size and one-digit industry.
- **Missing data.** Some specifications use surviving firms; others assign values to firms with missing growth outcomes according to exit reason.

Interpretation. Operating performance provides a non-stock-market check on whether mandated disclosure was associated with better firm behavior after compliance.

5 Methodology and empirical specifications

5.1 Conceptual framework

The paper uses a diversion-based governance view of disclosure. Managers or insiders may divert resources through perks, empire building, insider trading, or other activities that reduce outside shareholder value. Mandatory disclosure may raise the probability of detection or punishment.

- If disclosure reduces expected diversion, outside shareholders should expect higher future payouts.
- For firms already public before the law, this creates a one-time stock-price increase when the law becomes expected or enforced.
- After the one-time repricing, expected returns should return to normal.
- A positive stock-price effect is a necessary condition for welfare improvement, but not sufficient, because some shareholder gains may be transfers from insiders.

Interpretation. The empirical tests ask whether the market behaved as if disclosure rules made outside shareholders better off.

5.2 Firm-level registration event study

The first test studies weekly firm-level returns around the public announcement that a firm had begun complying.

- The dependent variable is the OTC firm’s weekly return minus the return of a matched NYSE/AMEX size/book-to-market cell.
- The event indicator equals one from eight weeks before to one week after the SEC News Digest announcement.

- The regression is estimated separately for 0–4 and 2–4 firms.
- Some specifications include the market factor, size factor, book-to-market factor, and momentum factor.

Main implication. Positive event-window coefficients show that investors rewarded firms when compliance became known.

5.3 Portfolio return model

For each OTC group and comparison group, the paper constructs equal-weighted portfolio returns:

$$R_{gt} = \frac{1}{N_{gt}} \sum_{i=1}^{N_{gt}} R_{igt}.$$

The basic four-factor model is:

$$R_{gt} - R_{ft} = \alpha_g + \beta_{1g}(R_{mt} - R_{ft}) + \beta_{2g}SMB_t + \beta_{3g}HML_t + \beta_{4g}MOM_t + \epsilon_{gt}.$$

- $R_{mt} - R_{ft}$ is the market excess return.
- SMB_t is the size factor.
- HML_t is the book-to-market factor.
- MOM_t is the momentum factor.
- α_g is the abnormal excess return for group g .

Interpretation. The factor model adjusts for standard risk and return patterns that differ across firm types.

5.4 Differenced OTC-minus-4–4 specification

The preferred portfolio test differences each OTC group return from its matched 4–4 comparison group:

$$R_{OTC,t} - R_{4-4,t} = \alpha + (\Delta\beta_1)(R_{mt} - R_{ft}) + (\Delta\beta_2)SMB_t + (\Delta\beta_3)HML_t + (\Delta\beta_4)MOM_t + u_t.$$

- The key coefficient is the average abnormal excess return α .
- The comparison removes shocks common to the OTC group and its matched listed firms.
- The paper reports unadjusted, CAPM, four-factor, buy-and-hold, value-weighted, and industry-matched variants.

Interpretation. If the 0–4 group earns positive abnormal returns relative to matched 4–4 firms in Period 1, this is evidence that the law raised the value of firms newly subject to disclosure.

5.5 Subperiod and Period 2 tests

The paper splits the main window and then tests a validation period.

- **Period 1a.** Tests whether stock prices reacted while the law was proposed, debated, and passed.
- **Period 1b.** Tests whether stock prices reacted when compliance uncertainty was resolved and filings became known.
- **Period 2.** Tests whether the same OTC groups continued to earn abnormal returns after the law’s information content had been resolved.

Interpretation. A true law-related treatment effect should appear in the information period and disappear in the later validation period.

5.6 Operating-performance regressions

The operating-performance specification compares growth from 1962 to 1966:

$$Y_i = a + b_1 \mathbf{1}\{0-4_i\} + b_2 \mathbf{1}\{2-4_i\} + \gamma \log(Assets_{i,1962}) + \text{Industry FE} + e_i.$$

- The omitted category is 4–4 listed firms.
- Outcomes are net income growth and sales growth.
- Positive coefficients for 0–4 firms imply that the most affected OTC firms improved more than unaffected listed firms.

Interpretation. This test examines whether the stock-market reaction was consistent with later improvements in operating performance.

6 Identification and interpretation

6.1 What the paper can identify

The paper identifies how shareholders valued the imposition of mandatory disclosure on firms that had avoided or lacked those requirements before 1964. The evidence is strongest for the “treatment on the untreated”: OTC firms that were not already voluntarily subject to the full disclosure regime.

- The study estimates the stock-market value of new disclosure requirements for affected existing firms.
- It does not directly estimate the effect of disclosure on firms that would voluntarily disclose anyway.
- It also does not fully estimate economy-wide welfare, because shareholder gains may include transfers from insiders.

6.2 Why ex ante group assignment matters

The OTC groups are defined using 1962 information rather than later realized compliance.

- Using realized compliance would mechanically use information not known at the beginning of Period 1.
- Firms may have manipulated assets or shareholder counts to avoid compliance; ex ante grouping captures the policy-relevant expected treatment.
- Any mismatch between ex ante classifications and market beliefs should generally attenuate differences across groups.

Interpretation. Ex ante assignment makes the design closer to a clean treatment-intensity test.

6.3 Why matched 4–4 firms matter

A simple comparison of OTC returns to zero is not enough because OTC firms differ systematically from listed firms.

- OTC firms are generally smaller and less liquid than listed firms.
- Size and book-to-market are known predictors of average returns.
- The matched 4–4 portfolios adjust for these observable return-relevant characteristics.
- The four-factor model further adjusts for market, size, value, and momentum exposure.

Interpretation. The strongest Period 1 estimates compare affected OTC firms to listed firms that look similar in size and book-to-market but were not affected by the law.

6.4 Why Period 2 matters

Period 2 is a validation period after the main disclosure-related news should already have been priced.

- If OTC firms simply outperform listed firms for unrelated reasons, abnormal returns could continue in Period 2.
- The paper finds little evidence of abnormal returns for any OTC group in Period 2.
- This supports the interpretation that Period 1 returns reflect the 1964 Amendments rather than a general OTC-return anomaly.

6.5 Stock returns versus welfare

The paper is careful about the welfare interpretation.

- Positive stock returns mean outside shareholders valued the disclosure requirements.

- Some shareholder gains may be transfers from insiders who lost private benefits of control or diversion opportunities.
- If diversion destroys resources, then reducing it can generate welfare gains.
- If diversion is purely redistributive, the stock-market gain may overstate social welfare gains.

Interpretation. The paper provides strong evidence of shareholder-value gains, but more limited evidence on total welfare.

6.6 External validity

The results are informative but not universal.

- The treated 0–4 firms are not a random sample. Many had avoided public offerings or exchange listing despite apparent size.
- The results may be most relevant for markets where disclosure standards are weaker than the post-1934 U.S. listed-firm regime.
- The results are less directly informative about more stringent modern rules, such as Sarbanes-Oxley, because the compliance costs and marginal information content may be different.

7 Tables and figures

7.1 Table I and Table II: Timeline and treatment groups

Read Table I:

- Table I lays out the research design calendar.
- Period 1 starts on January 1, 1963, before the SEC Special Study and the main legislative action.
- Key legislative events occur in 1963 and 1964: the SEC Special Study, Senate action, House action, passage, and President Johnson’s signature.
- Period 1b begins after passage and covers firm-level initial filing deadlines.
- Period 2 begins on November 15, 1965, after the key filing window had ended.

Read Table II:

- Table II summarizes the 0–4, 2–4, 0–0, 2–2, and 4–4 groups.
- The 0–4 group has 240 firms at the beginning of 1963, while the 2–4 group is much larger with 738 firms.
- Compliance with all four disclosure requirements rises sharply for 0–4 and 2–4 firms after the law.
- The 0–4 and 2–4 firms are much larger than 0–0 and 2–2 firms, consistent with the size thresholds in the law.

- 0–4 firms have higher bid-ask spreads than 2–4 firms, consistent with more information asymmetry among nonfilers.

Economic message: The timeline and group construction are the backbone of the empirical strategy. The law affected firms at different treatment intensities, and the key information events were concentrated in a clear window.

TABLE I
KEY DATES ASSOCIATED WITH THE SECURITIES ACTS AMENDMENTS
OF 1964 AND WITH RESEARCH DESIGN

1. Beginning of Period 1 (and Period 1a), January 1, 1963: sample of OTC firms' stock returns begins. We assume full impact of 1964 Securities Acts Amendments on stock returns occurs in this period.
2. April 3, 1963: the SEC releases the first part of the <i>Special Study</i> . It recommends imposing exchange disclosure rules on OTC securities.
3. July 9, 1963: Senate subcommittee unanimously approves bill extending disclosure requirements to all OTC firms.
4. July 30, 1963: Senate passes a bill extending disclosure rules to OTC firms with at least \$1 million in assets and 500 shareholders.
5. February, 1964: President Johnson endorses legislation and lobbies Congress to further its prospects for passage.
6. March 19, 1964: House subcommittee passes bill, and it becomes evident Congress as a whole will pass it.
7. August 5–6, 1964: House and Senate pass the bill.
8. End of Period 1a, August 24, 1964: President Johnson signs 1964 Securities Acts Amendments into law on August 20, 1964.
9. Beginning of Period 1b, August 24, 1964: OTC firms with at least \$1 million of assets and 750 shareholders required to initially file with SEC during this period.
10. April 30, 1965: deadline for covered firms with fiscal year ends between July 1, 1964, and December 31, 1964, to initially file. Covered firms with fiscal year ends between January 1, 1965, and June 30, 1965, required to initially file 120 days after fiscal year end.
11. October 31, 1965: last deadline for filing registration statement with SEC for newly covered firms.
12. End of Period 1 (and Period 1b), November 15, 1965: Period 1 ends two weeks after October 31 deadline to allow for gradual dissemination of news about which firms filed with the SEC.
13. Beginning of Period 2, November 15, 1965: virtually no new information about the law or which firms would comply with its requirements was revealed in this period. This period is used to assess the validity of the Period 1 findings.
14. End of Period 2, December 31, 1966: sample of OTC firms' stock returns ends.

Table I: Key dates associated with the 1964 Securities Acts Amendments and research design.

TABLE II
FIRM CHARACTERISTICS BY MANDATORY DISCLOSURE GROUPS, 1962

	0-4	2-4	0-0	2-2	4-4					
Panel A. Number of firms										
Week 1, 1963	240	738	124	94	1915					
Week 52, 1966	186	610	67	63	1668					
Survival rate	77.5	82.7	54.0	67.0	87.1					
Panel B. Percent of firms in week 1, 1963 sample that comply with all four mandatory disclosure requirements as of July of relevant year										
1963	9.2	4.7	4.0	0	100					
1964	12.1	16.4	5.7	4.3	100					
1965	74.6	69.7	22.6	23.4	100					
1966	74.6	80.5	28.2	35.1	100					
1967	80.8	82.4	37.1	54.3	100					
Panel C. Firm characteristics										
	N	Mean	N	Mean	N	Mean	N	Mean		
		(Median)		(Median)		(Median)		(Median)		
Information from <i>Barron's</i> /CRSP for 1963, week 1										
Share price	240	\$19.8	738	\$17.1	124	\$9.8	94	\$6.1	1915	\$25.0
		(\$12.3)		(\$12.9)		(\$4.4)		(\$4.4)		(\$19.4)
Bid-ask spread	240	11.1%	738	9.9%	124	16.7%	94	15.3%	na	na
(%)		(10.0%)		(9.1%)		(17.1%)		(14.9%)		na
Information from <i>Moody's</i> /COMPUSTAT for 1962										
# Shareholders	232	2946	695	4757	13	506	14	577	na	na
		(1861)		(2300)		(389)		(435)		na
Assets	240	\$44.5	738	\$33.4	100	\$2.7	89	\$3.0	1906	\$207.9
		(\$12.2)		(\$14.7)		(\$1.3)		(\$2.1)		(\$41.7)
Market capitalization	237	\$44.6	734	\$26.5	23	\$4.4	39	\$2.5	1915	\$189.1
		(\$9.1)		(\$10.4)		(\$1.9)		(\$2.3)		(\$28.5)
Book value of equity	230	\$25.5	727	\$15.3	19	\$3.5	27	\$2.1	1904	\$134.1
		(\$8.0)		(\$7.4)		(\$1.3)		(\$1.6)		(\$28.0)
Net income	238	\$2.7	731	\$1.6	99	\$0.1	89	\$0.2	1903	\$11.5
		(\$0.7)		(\$0.8)		(\$0.1)		(\$0.2)		(\$1.9)
Sales	223	\$45.9	718	\$32.3	99	\$3.3	86	\$4.6	1881	\$187.7
		(\$14.6)		(\$16.3)		(\$2.0)		(\$2.8)		(\$48.0)
Information from S&P Security Owners' Stock Guide										
% Held by institutions	120	4.04%	490	4.42%	8	2.52%	14	5.79%	na	na
		(2.19%)		(2.69%)		(1.33%)		(3.56%)		na

All firms tabulated in the OTC and NYSE/AMEX groups are present in their respective samples as of the first week of 1963. NYSE/AMEX firms tabulated exclude firms that enter from the *Barron's* sample during the 1963–1966 period. All dollar figures are in millions, except per share data. Means and medians are calculated using available observations.

Table II: Firm characteristics by mandatory-disclosure group.

7.2 Table III and Figure I: Firm-level SEC registration event study

Read Table III:

- Table III estimates abnormal returns around the ten-week event window surrounding SEC registration announcements.
- The event-window coefficient is about 0.35 percent per week for both 0–4 and 2–4 firms.
- Over ten weeks, this corresponds to roughly 3.5 percent cumulative abnormal excess returns.
- The estimates are stable with and without the four-factor adjustment.

- The result uses firm-level variation in the timing of compliance announcements.

Read Figure I:

- Figure I plots cumulative abnormal excess returns around the registration week.
- The lines are relatively flat before the event window.
- They rise during the event window and then flatten afterward.
- The visual evidence matches the Table III event-study result.

Economic message: Investors reacted positively when firms began complying. This is the most direct evidence that shareholders valued the new disclosure regime.

	0-4 - 4-4			2-4 - 4-4		
	(1)	(2)	(3)	(4)	(5)	(6)
1 (10-Week period from filing to registration)	0.358 (0.177)	0.350 (0.173)	0.344 (0.162)	0.369 (0.103)	0.350 (0.097)	0.312 (0.092)
Constant	0.002 (0.099)	0.082 (0.082)	0.060 (0.069)	-0.151 (0.090)	-0.040 (0.071)	-0.013 (0.051)
R ²	0.001	0.007	0.005	0.001	0.004	0.003
Factor model	None	4 Factors	4 Factors	None	4 Factors	4 Factors
Firms	145	145	215	417	417	693
Observations (firm-weeks)	8,845	8,845	12,824	25,857	25,857	42,277
Full sample	No	No	Yes	No	No	Yes

The sample in columns (6) and (3) includes all firms that belong to either group 0-4 or group 2-4 and for which returns during the period are available. Columns (1), (2), (4), and (5) limit the sample to firms that register with the SEC before the end of Period 1. The indicator variable 1 (16-week period from filing to registration) equals 1 between the 8 weeks before registration and the 1 week after registration (both inclusive). Columns (1) and (4) include a constant terms as the only other explanatory variable. Columns (2), (3), (5), and (6) also include the market, size, book-to-market, and momentum factors. Clustered standard errors, allowing for correlation across firms within weeks, are in parentheses.

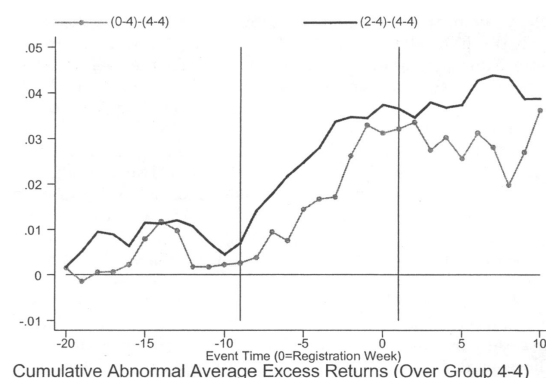


FIGURE I
Cumulative Abnormal Excess Returns Near SEC Registration
The vertical lines define a ten-week event window that starts eight weeks prior to the SEC News Digest announcement that a firm had registered under the 1964 Amendments and ends one week after this date. Abnormal returns are calculated based on the residuals from a firm-level regression that includes an intercept and four return factors (the excess return on the market portfolio and the size, book-to-market, and momentum factors).

Table III: Firm-level analysis of SEC registration.

Figure I: Cumulative abnormal excess returns near SEC registration.

7.3 Figures II and III: Raw cumulative returns and comparison groups

Read Figure II:

- Figure II compares cumulative returns of the four OTC portfolios and their listed comparison portfolios.
- In the OTC panel, 0-4 and 2-4 firms have relatively high cumulative returns over Period 1.
- In the NYSE/AMEX comparison panel, the matched listed portfolios move more similarly.
- This suggests that the cross-OTC-group return pattern is not simply a mechanical feature of matched listed firms.

Read Figure III:

- Figure III compares the 0-4 group and 2-4 group directly to their matched 4-4 comparison groups.

- The 0–4 group modestly outperforms its comparison portfolio in raw cumulative returns during Period 1.
- The 2–4 group looks much closer to its matched listed comparison group.
- The figure helps motivate why factor adjustment matters for the main 0–4 Period 1 estimate.

Economic message: The raw plots show that the most affected OTC firms did relatively well during the treatment window, but the strongest inference comes from the factor-adjusted and matched specifications.

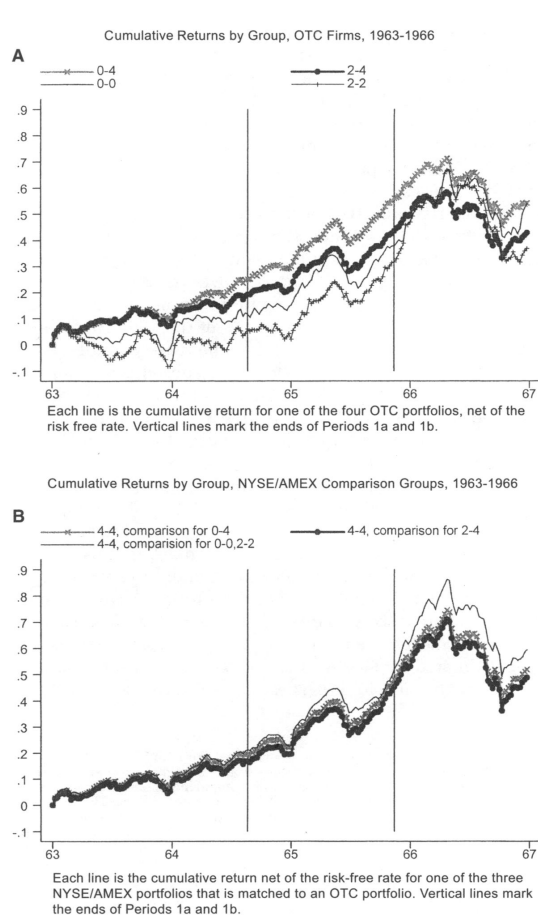


FIGURE II

Figure II: Cumulative returns by OTC group and NYSE/AMEX comparison groups.

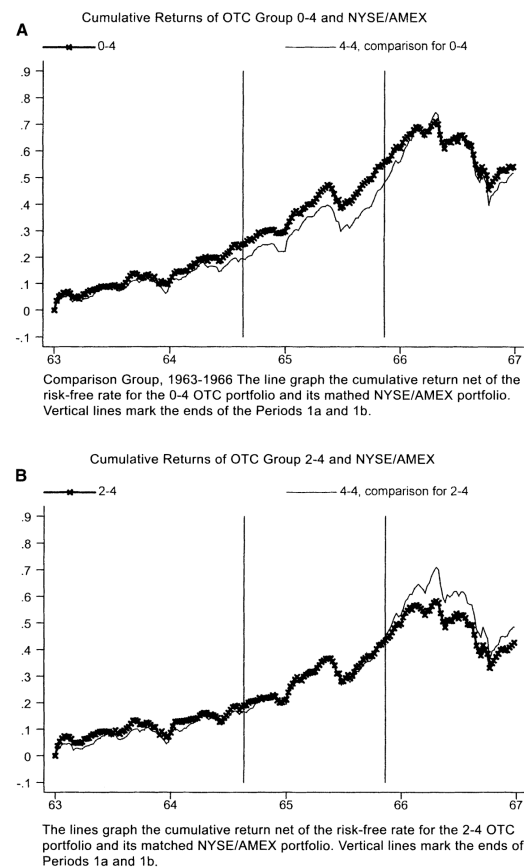


FIGURE III

Figure III: Cumulative returns of 0–4 and 2–4 groups versus matched NYSE/AMEX portfolios.

7.4 Tables IV and V: Period 1 abnormal returns and subperiods

Read Table IV:

- Table IV reports average abnormal excess returns over Period 1.
- In the preferred factor-adjusted OTC-minus-4–4 specifications, the 0–4 group has positive abnormal excess returns.

- The 0–4 estimates imply cumulative abnormal excess returns between about 11.5 and 22.1 percent.
- The 2–4 estimates are smaller and generally not statistically significant.
- The 0–0 and 2–2 estimates are imprecise and do not show a similar positive pattern.
- The 0–4 group also outperforms other OTC groups in many specifications.

Read Table V:

- Table V splits Period 1 into Period 1a and Period 1b.
- The 0–4 group has positive abnormal returns in both subperiods, with much of the effect appearing during the legislative debate and passage period.
- Period 1b estimates are positive in many specifications but are more sensitive.
- The 2–4 evidence is weaker and less stable.

Economic message: The core valuation effect is concentrated among 0–4 firms, the group most exposed to new disclosure. The market capitalized much of the value when the law became likely, before all firms had actually filed.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	0-4				(0-4)-(4-4)			
$\alpha_{0-4}^{Pd,1}$ (%)	0.378	0.187	0.053	0.078	0.097	0.129	0.083	0.149
	(0.077)	(0.050)	(0.048)	(0.048)	(0.046)	(0.048)	(0.048)	(0.047)
R^2	—	0.639	—	0.028	0.191	0.195	0.213	0.160
	2-4				(2-4)-(4-4)			
$\alpha_{2-4}^{Pd,1}$ (%)	0.296	0.095	-0.010	0.013	0.024	0.057	0.035	0.056
	(0.080)	(0.049)	(0.045)	(0.043)	(0.040)	(0.043)	(0.048)	(0.039)
R^2	—	0.667	—	0.026	0.133	0.120	0.166	0.079
	0-0				(0-0)-(4-4)			
$\alpha_{0-0}^{Pd,1}$ (%)	0.257	0.063	-0.095	-0.075	-0.095	-0.109	-0.116	0.000
	(0.110)	(0.092)	(0.096)	(0.095)	(0.098)	(0.117)	(0.135)	(0.105)
R^2	—	0.326	—	0.004	0.056	0.040	0.056	0.059
	2-2				(2-2)-(4-4)			
$\alpha_{2-2}^{Pd,1}$ (%)	0.214	0.033	-0.138	-0.131	-0.125	-0.071	-0.110	-0.021
	(0.130)	(0.115)	(0.106)	(0.107)	(0.114)	(0.116)	(0.127)	(0.112)
R^2	—	0.296	—	0.00	0.001	0.000	0.038	0.024
Test of equal α ,								
p -value								
$\alpha_{0-4}^{Pd,1} = \alpha_{2-4}^{Pd,1}$	0.014	0.015	0.064	0.054	0.053	0.046	0.284	0.020
$\alpha_{0-4}^{Pd,1} = \alpha_{0-0}^{Pd,1}$	0.149	0.167	0.100	0.087	0.049	0.051	0.146	0.160
$\alpha_{0-4}^{Pd,1} = \alpha_{2-2}^{Pd,1}$	0.097	0.128	0.054	0.032	0.030	0.062	0.107	0.100
Factor model	None	4F	None	CAPM	4F	4F	4F	4F
Buy and hold	No	No	No	No	No	Yes	Yes	No
Value weighted	No	No	No	No	No	No	Yes	No
Industry matched	No	No	No	No	No	No	No	Yes

The entries in columns (1) and (2) are the parameter estimates and heteroskedastic-consistent standard errors (in parentheses) on the constant from the estimation of versions of equation (2). Columns (3)–(8) are based on estimates of equation (3). Bold-face panel headings denote groups and the specification details are at the bottom of the table. The sample size is 148 weeks, but it is based on 149 calendar weeks. See footnote 24 for details. “CAPM” indicates that the regression controls for the return on the market. “4F” indicates controls for the market, size, book-to-market and momentum factors. Details on these factors, as well as our methodology for “buy and hold” returns, value weighting, and adjusting for industry returns are provided in the text.

Table IV: Average abnormal excess return in Period 1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	0-4				(0-4)-(4-4)			
$\alpha_{0-4}^{Pd,1a}$ (%)	0.299	0.223	0.066	0.100	0.115	0.141	0.122	0.165
	(0.095)	(0.076)	(0.064)	(0.065)	(0.068)	(0.070)	(0.074)	(0.068)
R^2	—	0.521	—	0.035	0.132	0.153	0.186	0.172
	0-4				(0-4)-(4-4)			
$\alpha_{0-4}^{Pd,1b}$ (%)	0.483	0.206	0.036	0.053	0.106	0.133	0.039	0.174
	(0.127)	(0.067)	(0.072)	(0.071)	(0.065)	(0.067)	(0.057)	(0.069)
R^2	—	0.778	—	0.021	0.328	0.281	0.252	0.194
	2-4				(2-4)-(4-4)			
$\alpha_{2-4}^{Pd,1a}$ (%)	0.230	0.129	0.030	0.052	0.052	0.086	0.141	0.088
	(0.102)	(0.081)	(0.061)	(0.059)	(0.066)	(0.067)	(0.070)	(0.065)
R^2	—	0.583	—	0.016	0.097	0.089	0.140	0.127
	2-4				(2-4)-(4-4)			
$\alpha_{2-4}^{Pd,1b}$ (%)	0.381	0.112	-0.062	-0.039	0.026	0.058	-0.068	0.061
	(0.128)	(0.058)	(0.065)	(0.063)	(0.053)	(0.060)	(0.064)	(0.049)
R^2	—	0.787	—	0.046	0.265	0.229	0.282	0.112
Test of equal α ,								
p -value								
$\alpha_{0-4}^{Pd,1a} = \alpha_{2-4}^{Pd,1a}$	0.133	0.070	0.426	0.278	0.214	0.258	0.773	0.149
$\alpha_{0-4}^{Pd,1a} = \alpha_{0-4}^{Pd,1b}$	0.045	0.091	0.057	0.079	0.153	0.173	0.053	0.060
Factor model	None	4F	None	CAPM	4F	4F	4F	4F
Buy and hold	No	No	No	No	No	Yes	Yes	No
Value weighted	No	No	No	No	No	No	Yes	No
Industry matched	No	No	No	No	No	No	No	Yes

The entries in columns (1) and (2) are the parameter estimates and heteroskedastic-consistent standard errors (in parentheses) on the constant from the estimation of versions of equation (2). Columns (3)–(8) are based on estimates of equation (3). Bold-face panel headings denote groups and the specification details are at the bottom of the table. The sample size is 84 weeks for Period 1a and 64 weeks for Period 1b. “CAPM” indicates that the regression controls for the return on the market. “4F” indicates controls for the market, size, book-to-market, and momentum factors. Details on these factors, as well as our methodology for “buy and hold” returns, value weighting, and adjusting for industry returns are provided in the text.

Table V: Period 1a and Period 1b abnormal excess returns.

7.5 Tables VI and VII: Period 2 validation and operating performance

Read Table VI:

- Table VI repeats the abnormal-return analysis for Period 2.
- None of the main OTC groups earns economically or statistically compelling abnormal excess returns in this later period.
- This supports the idea that Period 1 returns were tied to disclosure news rather than persistent OTC outperformance.

Read Table VII:

- Table VII studies operating performance from 1962 to 1966.
- 0–4 firms improve net income growth relative to listed 4–4 firms.
- 0–4 firms also have much higher sales growth than comparable listed firms.
- 2–4 firms show smaller improvements.
- The operating results are consistent with the stock-market evidence that the law improved shareholder-relevant behavior.

Economic message: The lack of Period 2 abnormal returns validates the timing of the stock-return evidence. The operating-performance improvements suggest the market reaction was not purely speculative.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	0-4		(0-4)-(4-4)					
$\alpha_{0-4}^{Pd,2}$ (%)	-0.031	0.076	-0.097	-0.149	-0.022	-0.020	-0.105	0.018
	(0.231)	(0.083)	(0.111)	(0.090)	(0.077)	(0.115)	(0.093)	(0.095)
R^2	—	0.912	—	0.373	0.669	0.586	0.666	0.395
	2-4		(2-4)-(4-4)					
$\alpha_{2-4}^{Pd,2}$ (%)	-0.014	0.060	-0.075	-0.127	-0.035	0.007	-0.034	-0.022
	(0.235)	(0.073)	(0.106)	(0.082)	(0.074)	(0.092)	(0.088)	(0.069)
R^2	—	0.938	—	0.409	0.627	0.602	0.765	0.416
	0-0		(0-0)-(4-4)					
$\alpha_{0-0}^{Pd,2}$ (%)	0.276	0.303	0.150	0.111	0.172	0.169	0.336	0.215
	(0.318)	(0.212)	(0.206)	(0.213)	(0.235)	(0.261)	(0.317)	(0.235)
R^2	—	0.634	—	0.059	0.085	0.082	0.131	0.115
	2-2		(2-2)-(4-4)					
$\alpha_{2-2}^{Pd,2}$ (%)	0.092	0.033	-0.035	-0.062	-0.098	-0.080	-0.041	-0.054
	(0.334)	(0.160)	(0.180)	(0.186)	(0.166)	(0.199)	(0.237)	(0.150)
R^2	—	0.748	—	0.038	0.062	0.040	0.166	0.201
Test of equal α ,								
p -value								
$\alpha_{0-4}^{Pd,2} = \alpha_{2-4}^{Pd,2}$	0.765	0.801	0.703	0.708	0.841	0.738	0.404	0.619
$\alpha_{0-4}^{Pd,2} = \alpha_{0-0}^{Pd,2}$	0.117	0.281	0.213	0.214	0.339	0.432	0.154	0.395
$\alpha_{0-4}^{Pd,2} = \alpha_{2-2}^{Pd,2}$	0.532	0.803	0.751	0.668	0.680	0.787	0.797	0.697
Factor model	None	4F	None	CAPM	4F	4F	4F	4F
Buy and hold	No	No	No	No	No	Yes	Yes	No
Value weighted	No	No	No	No	No	No	Yes	No
Industry matched	No	No	No	No	No	No	No	Yes

The entries in columns (1) and (2) are the parameter estimates and heteroskedastic-consistent standard errors (in parentheses) on the constant from the estimation of versions of equation (2). Columns (3)–(8) are based on estimates of equation (3). Bold-face panel headings denote groups and the specification details are at the bottom of the table. The sample size is 58 weeks. “CAPM” indicates that the regression controls for the return on the market. “4F” indicates controls for the market, size, book-to-market, and momentum factors. Details on these factors, as well as our methodology for “buy and hold” returns, value weighting, and adjusting for industry returns are provided in the text.

Table VI: Average abnormal excess returns in Period 2.

	(1)	(2)	(3)	(4)
Dependent variable	Net income growth		Sales growth	
Sample	Survivors	Assigned	Survivors	Assigned
0–4 Indicator	0.038	0.030	1.091	0.835
	(0.015)	(0.012)	(0.149)	(0.121)
2–4 Indicator	0.013	0.012	0.171	0.161
	(0.009)	(0.008)	(0.091)	(0.077)
ln (Assets 1962)	-0.002	-0.001	-0.104	-0.089
	(0.003)	(0.002)	(0.025)	(0.022)
R^2	0.020	0.019	0.063	0.049
N	2408	2799	2331	2741

The entries in each column report the results from separate regressions. In columns (1) and (2) the dependent variable is “Net Income Growth,” defined as (Net Income in 1966 – Net Income in 1962)/Market Capitalization in 1962. In columns (3) and (4) the dependent variable is “Sales Growth,” and it is equal to (Sales in 1966 – Sales in 1962)/Sales in 1962. The first three sets of entries are the parameter estimates and standard errors (in parentheses) for the variables listed in the row headings. Sample is based on 0–4, 2–4, and 4–4 firms as of January 1963. The excluded category is 4–4 firms. In columns (1) and (3), the sample is restricted to those firms for which comparable 1966 information was available. In columns (2) and (4) firms with missing growth information were assigned the tenth or fiftieth percentile of firms in its group (see text for further details). The top and bottom 1 percent of observations in terms of the dependent variable are dropped. Regressions include one-digit SIC dummies. The sample means of the dependent variable in columns (1)–(4) are 0.093, 0.091, 0.863, and 0.857, respectively.

Table VII: Change in operating performance from 1962 to 1966.

7.6 Appendix tables: Exit treatment and factor loadings

Read Appendix 1:

- Appendix 1 summarizes how the authors treat OTC firms that move to NYSE/AMEX, change names, merge, liquidate, or disappear from Barron's.
- Exit returns are assigned where possible, which reduces bias from disappearing firms.
- This appendix is important because survivorship bias could otherwise contaminate OTC return estimates.

Read Appendix 2:

- Appendix 2 reports the factor loadings for OTC groups and matched NYSE/AMEX comparison groups in Periods 1 and 2.
- OTC and listed comparison groups have different market, size, book-to-market, and momentum exposures.
- These differences explain why the factor-adjusted analysis is important.

Economic message: The appendix evidence supports the credibility of the return construction and the factor-adjustment approach.

APPENDIX 1: TREATMENT OF FIRM CHANGES FOR OTC FIRMS					
	Jan 63	Jan 64	Jan 65	Jan 66	Dec 66
Sample size	1196	1129	1053	986	941
(a) Move to NYSE/AMEX	87	83	65	39	
Continued	87	83	65	39	
(b) Name change	9	15	10	7	
Continued	6	10	6	6	
Not continued (ER assigned/not assigned)	3 (1/2)	5 (3/2)	4 (3/1)	1 (0/1)	
(c) Merger	30	38	36	25	
Continued (merger week return ass/not ass.)	23 (20/3)	28 (20/8)	24 (19/5)	18 (16/2)	
Not continued (merger week return ass/not ass.)	7 (4/3)	10 (8/2)	12 (9/3)	7 (6/1)	
(d) Liquidations	5	5	5	5	
Not continued (ER assigned/not assigned)	5 (3/2)	5 (1/4)	5 (3/2)	5 (3/2)	
(e) Exit from Barron's, last price > \$2	29	14	14	13	
Not continued (ER assigned/not assigned)	29 (28/1)	14 (13/1)	14 (10/4)	13 (12/1)	
(f) Exit from Barron's, last price ≤ \$2	39	22	16	17	
Not continued (ER assigned/not assigned)	39 (31/8)	22 (19/3)	16 (13/3)	17 (11/6)	
Total exits/changes	198	177	138	106	
Total series continued	116	121	95	63	

"ER" stands for "exit return." Among the firms with discontinued time series of returns, "ER assigned"/"not assigned" refers to firms for which it was possible/impossible to assign exit returns within 20 weeks of the last Barron's price. Overall, we succeeded in assigning an exit return for 78 percent of the 233 firms where the time series of returns was not continued. The number of observations at the beginning of the following year does not exactly equal the number of observations at the beginning of the current year, minus the number of exits/changes, plus the number of series that are continued. This is because an exit return may only be assigned in the following calendar year. Furthermore, series that are continued may again be subject to one of the changes in (a)-(f). Such changes are not included in the above counts.

APPENDIX 2: FACTOR LOADINGS BY GROUP AND PERIOD								
	Pd 1	Pd 2	Pd 1	Pd 2	Pd 1	Pd 2	Pd 1	Pd 2
	0-4		2-4		0-0		2-2	
Market factor beta	0.701 (0.056)	0.866 (0.061)	0.750 (0.057)	0.837 (0.057)	0.588 (0.106)	0.815 (0.147)	0.720 (0.131)	0.783 (0.109)
Size factor beta	0.651 (0.117)	0.904 (0.133)	0.725 (0.117)	0.892 (0.095)	0.868 (0.186)	1.209 (0.259)	0.977 (0.204)	1.294 (0.193)
Book/market factor beta	0.014 (0.085)	0.236 (0.059)	0.070 (0.081)	0.282 (0.700)	0.037 (0.146)	0.345 (0.282)	0.205 (0.177)	0.565 (0.321)
Momentum factor beta	0.054 (0.074)	-0.248 (0.084)	-0.011 (0.085)	-0.170 (0.060)	0.159 (0.139)	-0.187 (0.198)	-0.200 (0.175)	-0.028 (0.170)
R ²	0.639	0.912	0.667	0.938	0.326	0.634	0.296	0.748
NYSE/AMEX comparison groups								
	For 0-4		For 2-4		For 0-0 and 2-2			
Market factor beta	0.840 (0.039)	0.982 (0.038)	0.868 (0.037)	0.992 (0.038)	0.747 (0.070)	0.944 (0.062)		
Size factor beta	0.969 (0.062)	1.087 (0.079)	0.985 (0.061)	1.089 (0.082)	1.039 (0.107)	1.357 (0.141)		
Book/market factor beta	0.250 (0.062)	0.268 (0.060)	0.214 (0.063)	0.245 (0.058)	0.245 (0.104)	0.341 (0.095)		
Momentum factor beta	-0.104 (0.070)	-0.067 (0.053)	-0.115 (0.068)	-0.063 (0.052)	-0.208 (0.126)	-0.123 (0.098)		
R ²	0.875	0.974	0.886	0.974	0.680	0.925		

The entries are the factor loadings (betas) on the four factors in the estimation of equation (2). Heteroskedastic-consistent standard errors are in parentheses.

Appendix 1: Treatment of firm changes for OTC firms.

Appendix 2: Factor loadings by group and period.

8 Short summary

- The paper studies the 1964 Securities Acts Amendments, which extended mandatory disclosure rules to large OTC firms.
- The main treated firms are 0–4 OTC firms: firms expected to move from no disclosure requirements to all four major requirements.
- The authors hand-collect weekly OTC returns from Barron’s and other historical sources because OTC firms were not in CRSP during the sample period.
- The empirical strategy compares affected OTC firms with less affected OTC firms and with matched NYSE/AMEX firms that were already subject to full disclosure.
- Around firm-level SEC registration announcements, 0–4 and 2–4 firms earn roughly 3.5 percent cumulative abnormal excess returns.
- Over Period 1, the most affected 0–4 firms earn abnormal excess returns of about 11.5 to 22.1 percent in the preferred matched and factor-adjusted specifications.
- Period 2 produces no similar abnormal returns, supporting the timing-based interpretation.
- Operating performance improves more for the most affected OTC firms than for unaffected listed firms.
- The results suggest that mandatory disclosure increased outside shareholder value, likely by reducing agency problems or insider diversion.
- The welfare interpretation is more cautious because some shareholder gains could be transfers from insiders rather than pure efficiency gains.

9 Conclusion

9.1 Core conclusion

Greenstone, Oyer, and Vissing-Jorgensen show that investors valued the imposition of mandatory disclosure rules on previously less regulated OTC firms. The strongest evidence comes from the firms that moved from zero to four disclosure requirements. These firms had positive abnormal returns during the law’s proposal and implementation window, and they also improved operating performance afterward.

9.2 Economic intuition

Mandatory disclosure changes the information and governance environment. More disclosure can make it harder for insiders to divert resources, hide poor performance, avoid shareholder scrutiny, or exploit private information. When the expected value of diversion falls, outside shareholders expect to receive more of the firm’s value, so stock prices rise.

9.3 Incremental contribution revisited

The paper makes the debate over disclosure regulation more concrete. Instead of relying on broad cross-country correlations or general arguments about investor protection, it studies a major U.S. disclosure shock with clear treatment intensity. The event study, portfolio-return tests, placebo-style Period 2 test, and operating-performance evidence all point in the same direction.

9.4 Implications

The findings imply that disclosure regulation can have real economic value, especially where insiders previously had wide scope to avoid external scrutiny. They also suggest that policy debates should distinguish between marginal costs for already transparent firms and marginal benefits for firms that are far from transparent.

9.5 Limitations

The main portfolio evidence for the 0–4 group relative to matched 4–4 firms relies on the validity of the standard factor model. The treated OTC firms are not random firms; they are firms that had not voluntarily subjected themselves to full disclosure. The estimates therefore speak most directly to the value of disclosure for firms that had avoided it. Finally, shareholder-value gains do not automatically equal welfare gains.

9.6 Takeaway

The enduring takeaway is that disclosure rules can matter because they discipline insiders. In this setting, investors appear to have viewed mandatory disclosure as valuable, and the firms most affected by the law experienced both positive stock-market reactions and subsequent operating improvements.